Hand Gesture Detection using Spatio-Temporal Graph Convolutional Networks on MediaPipe Skeletal Data

Mikhail Orekhov
RTU MIREA, Moscow, Russia
orekhov.m.v@edu.mirea.ru

Abstract

This paper presents a robust pipeline for binary dynamic gesture recognition using skeletal data. Using MediaPipe for keypoint extraction and a Spatio-Temporal Graph Convolutional Network (ST-GCN) for classification, we shift the focus from raw pixel data to geometric motion dynamics, ensuring resilience to environmental noise. Our approach utilizes a sliding window mechanism to achieve stable real-time inference. Experimental results on a manually collected dataset demonstrate a validation accuracy of 83% with a high precision threshold optimized for security-oriented applications. The proposed system offers an efficient solution for gesture-based authentication and human-computer interaction. Our implementation is available at https://github.com/UraniumSlashBroomer/Project-L.

1 Introduction

Dynamic gesture recognition is a pivotal task in sign language interpretation [1], VR/AR systems [3, 4], and human-computer interaction. Conventional convolutional neural network (CNN) architectures often lack robustness in these scenarios due to their sensitivity to environmental factors such as background noise and illumination changes. Shifting the focus from raw pixel data to skeletal keypoint sequences enables the model to capture the underlying motion dynamics more effectively. In this work, we present a pipeline that leverages MediaPipe [2] for keypoint extraction and a Spatio-Temporal Graph Convolutional Network (ST-GCN) [5] for binary gesture classification (Fig. 1). This approach demonstrates great potential for secure applications, such as user authentication via unique gesture recognition.

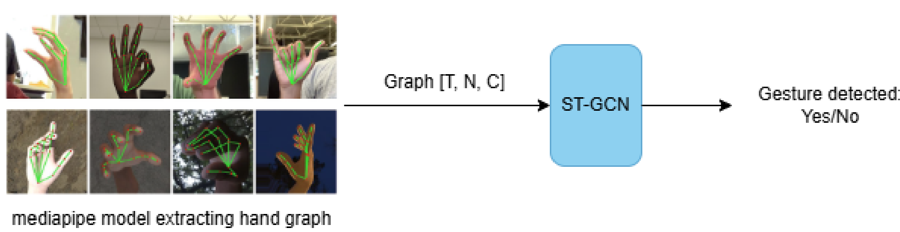

Figure 1: Overall pipeline of the proposed system

2 Method

2.1 Dataset and Data Preprocessing

We collected a dataset of 768 samples (balanced classes) using MediaPipe hand keypoints. Each sample is a tensor of shape [T×N×C], where T=30, N=21, C=3. Coordinates were normalized relative to the palm root and scaled by hand size. Horizontal flipping (p=0.5) was used for augmentation.

2.2 Classification model architecture

We use a 3-layer ST-GCN backbone followed by global average pooling and a linear classifier. Each block consists of spatial graph convolution and temporal convolution with ReLU and BatchNorm (Fig. 2).

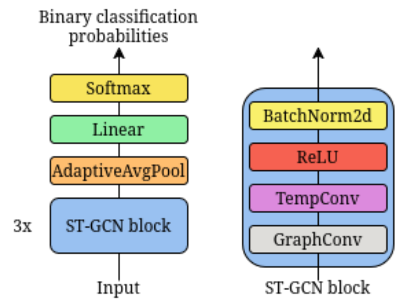

Figure 2: ST-GCN backbone architecture

2.3 Implementation details

We optimized the binary cross-entropy loss function (1) using the Adam optimizer.

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right] \tag{1}$$

During inference, a sliding window mechanism was implemented to maintain a buffer of 30 frames, with the classifier being triggered every 5 frames (stride). This optimization significantly reduces computational overhead while maintaining smooth, real-time performance.

3 Experiments and Results

The model uses channel sizes 64, 128, 196, total $\approx$ 565K parameters. Training is performed for 250 epochs with Adam (lr=3e-4).

The proposed system achieved a validation accuracy of 83% and a training accuracy of 85% (Fig. 3). Based on the ROC curve (Fig. 4) and confusion matrix (Fig. 5), a classification threshold of 0.7 was selected, yielding a True Positive Rate (TPR) of $\approx 87\%$ and a False Positive Rate (FPR) of $\approx 17\%$. In security-critical applications, such as gesture-based authentication, high precision is prioritized to minimize false acceptances and prevent unauthorized access.

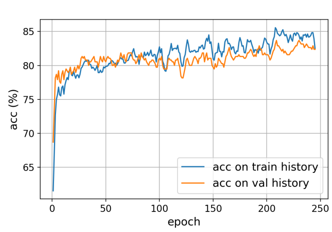

Figure 3: Accuracy

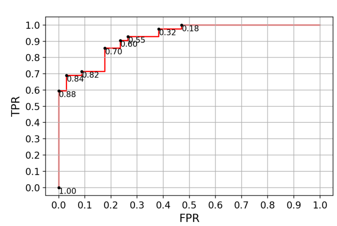

Figure 4: ROC Curve

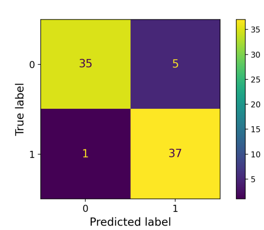

Figure 5: Confusion Matrix

4 Conclusion

We presented a skeletal-based ST-GCN pipeline for gesture recognition, achieving 83% accuracy with real-time performance. Future work includes multi-class extension and attention mechanisms.

References

[1] Brandone Fonya and Clarence Worrell. Real-time sign language gestures to speech transcription using deep learning, 2026.

[2] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. Mediapipe: A framework for building perception pipelines. *CoRR*, abs/1906.08172, 2019.

[3] Sandeep Reddy Sabbella, Sara Kaszuba, Francesco Leotta, Pascal Serrarens, and Daniele Nardi. Evaluating gesture recognition in virtual reality, 2024.

[4] Yikan Wang. Accessible gesture-driven augmented reality interaction system, 2025.

[5] Bing Yu, Haoteng Yin, and Zhanxing Zhu. Spatio-temporal graph convolutional neural network: A deep learning framework for traffic forecasting. *CoRR*, abs/1709.04875, 2017.